

1 List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.

ANSWER ::

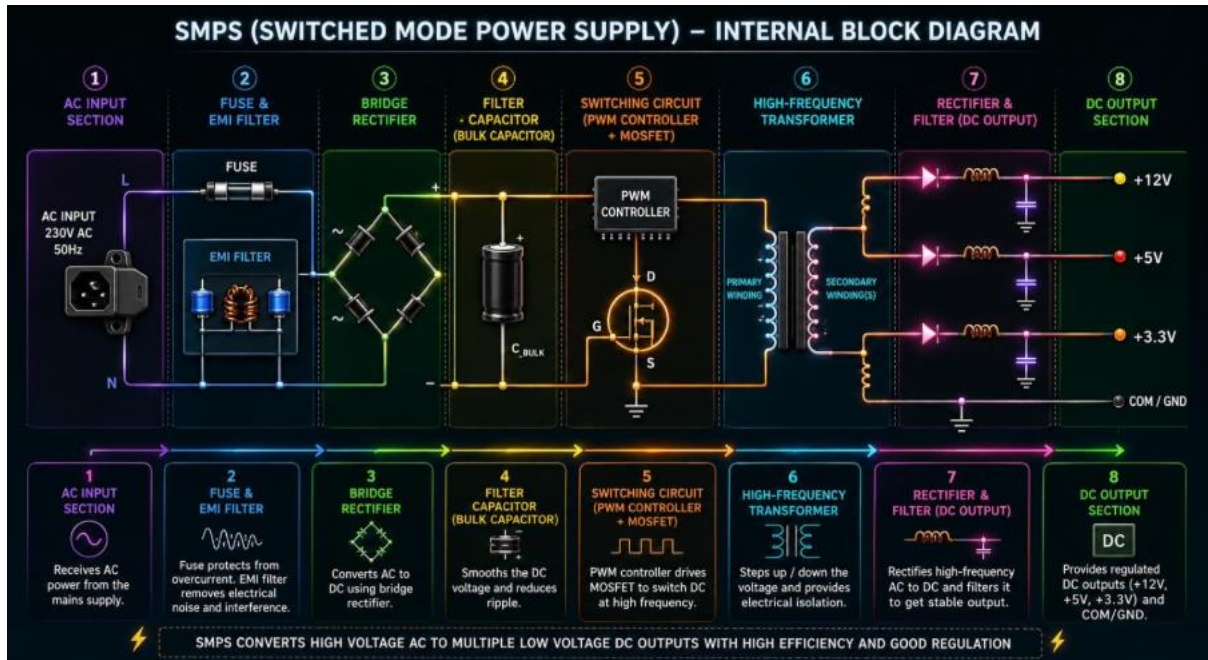
➤ **Devices that use SMPS (Switched Mode Power Supply)**

- ❑ **DESKTOP COMPUTER (CPU) ::** Uses an SMPS to convert AC power into DC power for the computer parts.
- ❑ **LAPTOP CHARGER ::** Uses an SMPS to provide safe and stable power for charging the laptop.
- ❑ **LED TELEVISION ::** Uses an SMPS to supply power efficiently and reduce heat.

❖ **DEVICES THAT USE UPS (Uninterruptible Power Supply)**

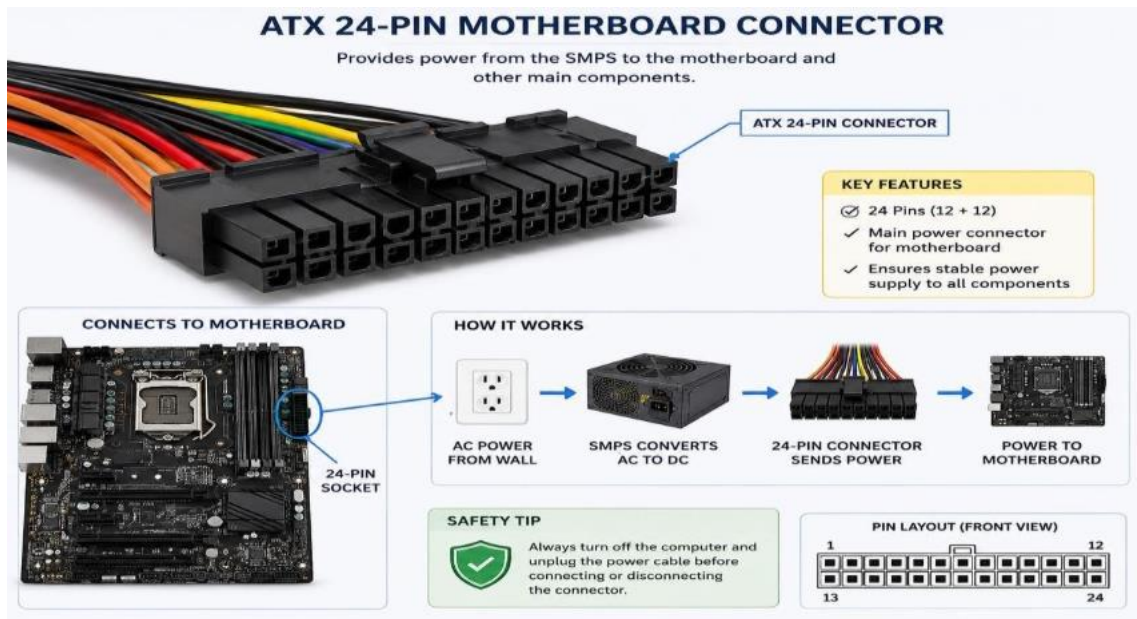
- **CCTV SECURITY SYSTEM ::** Uses a UPS to keep recording even during a power cut.
- **DESKTOP COMPUTER ::** Uses a UPS to keep recording even during a power cut.

2. Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections.



3. Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.

**ANSWER::**



### 1. ATX 24-pin Motherboard Connector ::

- **Name:** ATX 24-pin Motherboard Connector
- **Devices it powers:** It gives power to the motherboard.
- **Safety Tip:** Turn off the computer before connecting or removing the connector.

## 2. SATA Power Connector



**Name:**  
SATA Power Connector



**Devices it powers:**  
It powers SATA hard drives (HDD), SSDs, and DVD/Blu-ray drives.



**Safety Tip:**  
Connect it in the correct direction and never force the connector.

### Where it connects



### Connect in the correct direction



Never force the connector. ⚠

## 2. SATA Power Connector

- **Name:** SATA Power Connector.
- **Devices it powers:** It gives power to HDDs, SSDs, and DVD drives.
- **Safety Tip:** Connect it carefully in the correct direction. Do not force it.

# MOLEX CONNECTOR

## 4-Pin Peripheral Power Connector

### 1. WHAT IS MOLEX CONNECTOR?



Molex connector is a 4-pin peripheral power connector used to supply power to many internal components of a computer.

### 2. CONNECTOR - MALE (PLUG)



Pin 1  
↓  
+12V

Pin 2  
↓  
GND

Pin 3  
↓  
GND

Pin 4  
↓  
+5V

### 3. CONNECTOR - FEMALE (SOCKET)



Pin 1  
↓  
+12V

Pin 2  
↓  
GND

Pin 3  
↓  
GND

Pin 4  
↓  
+5V

### 4. HOW IT CONNECTS (ANIMATION STEPS)

1 Align the male connector with the female socket.



2 Insert the male connector into the female socket.



3 Push it firmly until it locks in place.



CLICK!

4 Connection secure! Power is supplied.



### 5. COMMON USES



HDD / Hard Drive



DVD Drive



Case Fan



Old Graphics Card

### 6. PINOUT SUMMARY



| PIN | WIRE COLOR | VOLTAGE | FUNCTION            |
|-----|------------|---------|---------------------|
| 1   | Yellow     | +12V    | Provides +12V Power |
| 2   | Black      | GND     | Ground              |
| 3   | Black      | GND     | Ground              |
| 4   | Red        | +5V     | Provides +5V Power  |

### 3. MOLEX CONNECTOR ::

- **Name:** Molex (4-pin) Connector.
- **Devices it powers:** It powers older IDE hard drives, optical drives, cooling fans, and some accessories.
- **Safety Tip:** Hold the plastic connector while removing it. Do not pull the wires.

## PCIe 6-PIN & 8-PIN POWER CONNECTOR FOR GRAPHICS CARDS (GPU)

**PCIe 6-PIN CONNECTOR**

**PINOUT (6-PIN)**

| PIN | SIGNAL |
|-----|--------|
| 1   | +12V   |
| 2   | +12V   |
| 3   | +12V   |
| 4   | GND    |
| 5   | GND    |
| 6   | GND    |

■ YELLOW = +12V  
■ BLACK = GND

**HOW TO CONNECT (ANIMATION STEPS)**

- 1 Align the 6-pin connector with the GPU socket.
- 2 Insert the connector straight into the GPU socket.
- 3 Push firmly until the latch clicks into place. CLICK!
- 4 Connection secure! Power is supplied to the GPU. ✓

**PCIe 8-PIN CONNECTOR**

**PINOUT (8-PIN)**

| PIN | SIGNAL |
|-----|--------|
| 1   | +12V   |
| 2   | +12V   |
| 3   | +12V   |
| 4   | +12V   |
| 5   | GND    |
| 6   | GND    |
| 7   | GND    |
| 8   | GND    |

■ YELLOW = +12V  
■ BLACK = GND

**HOW TO CONNECT (ANIMATION STEPS)**

- 1 Align the 8-pin connector with the GPU socket.
- 2 Insert the connector straight into the GPU socket.
- 3 Push firmly until the latch clicks into place. CLICK!
- 4 Connection secure! Power is supplied to the GPU. ✓

**COMMON USES**

Mid-Range GPU (6-Pin)      High-End GPU (8-Pin)

**KEY FEATURES**

- ✓ Provides stable +12V power to GPU.
- ✓ 6-Pin = Up to 75W
- ✓ 8-Pin = Up to 150W
- ✓ Always connect firmly for stable performance.

**CABLE EXAMPLE**

6-PIN PCIe CABLE      8-PIN PCIe CABLE

**NOTE**

⚠ Do not force the connector. Make sure it is fully inserted.

Use the correct connector for your GPU.

#### 4. PCIe 6/8 PIN CONNECTOR ::

- **Name:** PCIe 6-pin / 8-pin GPU Power Connector.
- **Devices it powers:** It gives extra power to the **graphics card (GPU)**.
- **Safety Tip:** Insert it properly until the **clip clicks**. Do not force it.

4. Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.

ANSWER ::

1. **Turn off the PC:** Switch off the PC and unplug the main power cable.
2. **Connect the Motherboard:** Connect the SMPS to the motherboard using the **24-pin ATX cable**.
  - **Cable Name:** ATX Main Power Cable
  - **Connector Name:** 24-pin ATX Connector
3. **Connect the CPU:** Connect the CPU power using the **4/8-pin EPS cable**.
  - **Cable Name:** PCIe Power Cable
  - **Connector Name:** 6-pin / 8-pin PCIe Connector
4. **Connect the Graphics Card:** Connect the GPU using the **PCIe 6/8-pin cable**.
  - **Cable Name:** PCIe Power Cable
  - **Connector Name:** 6-pin / 8-pin PCIe Connector
5. **Connect Storage Devices:** Connect the HDD/SSD using the **SATA power cable**.
  - **Cable Name:** SATA Power Cable
  - **Connector Name:** 15-pin SATA Power Connector
6. **Check all connections:** Make sure all connectors are properly inserted and locked.

7. **Turn on the PC:** After checking everything, connect the main power cable and turn on the PC.